

- complex numbers: $(a, b) \in \mathbb{R} \Rightarrow a + bi$
- 1.1 $(a+bi) + (c+di) = (a+c) + (b+d)i$ $(a+bi)(c+di) = (ac-bd) + (ad+bc)i$ $a, b, c, d \in \mathbb{R}$ **UNIQUE**
list of length $n \Rightarrow$ ordered collection of n elements
- 1.8 two lists equal iff same length & same elements in same order
- 1.11 $\mathbb{F}^n = \{(x_1, \dots, x_n) : x_k \in \mathbb{F} \text{ for } k=1, \dots, n\}$ x_k : k^{th} coordinate
- * Field: set containing at least 2 distinct elements 0 and 1 with addition/multiplication
- 1.19 addition: $u+v \in V$ $u, v \in V$ scalar multiplication: $\lambda v \in V$ $\lambda \in \mathbb{F}$ $v \in V$
- 1.20 vector space: commutativity, associativity, additive/multiplicative identity, additive inverse, distributive properties
- 1.24 $\mathbb{F}^S$: if S is a set, $\mathbb{F}^S$ set of functions from S to $\mathbb{F}$
- 1.33 subspace: vector space w/ same additive identity, addition, scalar multiplication on V
sum of subspaces: $V_1 + \dots + V_m = \{v_1 + \dots + v_m : v_i \in V_i, \dots, v_m \in V_m\}$ analogous union subsets
- 1.41 Direct Sum: $V_1 \oplus \dots \oplus V_m = V_1 + \dots + V_m$ where each element of $V_1 + \dots + V_m$ can be written in only one way as a sum $v_1 + \dots + v_m$, $v_k \in V_k$
- 1.45 Direct Sum iff only way to write 0 as a sum $v_1 + \dots + v_m$ is by taking each $v_k = 0$
- 1.46 two subspaces: $U+W$ is direct sum $\Leftrightarrow U \cap W = \{0\}$
- 2.2 lin. combo of $v_1, \dots, v_m \in V$: $a_1 v_1 + \dots + a_m v_m$, $a_1, \dots, a_m \in \mathbb{F}$
- 2.4 $\text{span}(v_1, \dots, v_m) = \{a_1 v_1 + \dots + a_m v_m : a_1, \dots, a_m \in \mathbb{F}\}$ span of empty list is $\{0\}$
- 2.10 polynomial w/ coefficients in $\mathbb{F}$ if there is $a_0, \dots, a_m$ s.t. $p(z) = a_0 + a_1 z + \dots + a_m z^m$
- 2.12 $P_n(\mathbb{F})$ set of polynomials degree at most n
- 2.15 lin. indep. iff $a_1 v_1 + \dots + a_m v_m = 0$ when $a_1 = \dots = a_m = 0$; empty list
- 2.17 lin. dep. iff not all a 's need to equal zero for $a_1 v_1 + \dots + a_m v_m = 0$
- 2.19 if $v_1, \dots, v_m$ is lin. dep. $\Rightarrow$ there exists k s.t. $v_k \in \text{span}(v_1, \dots, v_{k-1})$ if v_k removed, span stays same
- 2.22 length of LI list $\leq$ length spanning list
- 2.26 basis: list of vectors in V that is lin. indep and spans V $\nearrow a_1, \dots, a_n \in \mathbb{F}$
- 2.28 list $v_1, \dots, v_n$ in V iff every $v \in V$ can be written uniquely as $v = a_1 v_1 + \dots + a_n v_n$
- 2.30 every spanning list can be reduced to a basis vector space
- 2.32 every lin. indep. list extends to a basis
- 2.33 every subspace of V is part of a direct sum equal to V
- 2.34 basis length doesn't depend on basis 2.35 $\dim V \Rightarrow$ length of any basis
- 2.37 if V is f.d. and U is a subspace of V , then $\dim U \leq \dim V$
- 2.38 lin. ind. of right length is a basis of V 2.39 subspace of full dimension = whole space
- 2.42 If V is f.d. every spanning list in V of length $\dim V$ is a basis of V
- 2.43 If V_1 & V_2 are subspaces of f.d. vector space: $\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$
- 3.1 linear map $T: V \rightarrow W$: additivity, homogeneity 3.2 $\mathcal{L}(V, W) \rightarrow V \times W$ $\mathcal{L}(V) = \mathcal{L}(V, V) \rightarrow V \times V$
- 3.4 Given $v_1, \dots, v_n$ basis of V & $w_1, \dots, w_n \in W$, then $T: V \rightarrow W$ s.t. $Tv_k = w_k \forall k=1, \dots, n$
- 3.7 product of linear maps: $T \in \mathcal{L}(U, V)$ $S \in \mathcal{L}(V, W)$ $ST \in \mathcal{L}(U, W) \Rightarrow (ST)(u) = S(Tu)$
associativity, identity, distributive properties
- 3.10 L.M. take 0 to 0 $T \in \mathcal{L}(V, W)$ $T(0) = 0$ 3.14 Injective if $Tu = Tv \Rightarrow u = v$
- 3.11 null $T = \{v \in V : Tv = 0\}$, is a subspace of V 3.15 T injective iff $\text{null } T = \{0\}$
- 3.16 range $T = \{Tv : v \in V\}$ 3.21 Fundamental Thm of Linear maps: V f.d. and $T \in \mathcal{L}(V, W)$
- 3.19 $T: V \rightarrow W$ surjective if $\text{range} = W$ • range T f.d. • $\dim V = \dim \text{null } T + \dim \text{range } T$
- 3.22 $\dim V > \dim W \Rightarrow \nexists$ injective $T: V \rightarrow W$ 3.23 $\dim V < \dim W \Rightarrow \nexists$ surjective $T: V \rightarrow W$
- 3.26 more variables than equations $\Rightarrow$ non zero slns 3.28 inhomogeneous $\Rightarrow$ more eqns than variables $\Rightarrow$ no sln
- 3.29 A is $m \times n$ 3.31 $M(T) = Tv_k = A_{1k} w_1 + \dots + A_{mk} w_m$ 3.35 $M(S+T) = M(S) + M(T)$ 3.38 $M(\lambda T) = \lambda M(T)$
- 3.39 $\mathbb{F}^{m \times n} \Rightarrow$ all $m \times n$ matrices w/ entries in $\mathbb{F}$ 3.40 $\dim \mathbb{F}^{m \times n} = mn$ 3.41 $(AB)_{j,k} = \sum_{i=1}^n A_{ji} B_{ik}$
- 3.43 $T \in \mathcal{L}(U, V)$ $S \in \mathcal{L}(V, W) \Rightarrow M(ST) = M(S) \cdot M(T)$ 3.48 $(AB)_{j,k} = A_{j,i} B_{i,k}$ 3.50 Ab for $b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \Rightarrow b_1 A_{1j} + \dots + b_n A_{nj}$
- 3.51 $C^{m \times n}, R^{m \times n}$: col k of $CR =$ l.c. col C w/ coeff from col k of R row j of $CR =$ l.c. rows R w/ coeff from row j of C
- 3.52 col rank: dimension of span of cols of A row rank: dimension of span of rows of A $\nearrow$ s is inverse
- 3.56 col-row factorization: $A = CR$ 3.57 col rank = row rank 3.59 invertible $\Leftrightarrow ST = I$ and $TS = I$
- 3.61 T invertible: $TT^{-1} = T^{-1}T = I$ 3.63 invertible $\Leftrightarrow$ injective and surjective $\nexists$ inverse is unique
- 3.65 if $\dim V = \dim W$: invertible $\Leftrightarrow$ injective $\Leftrightarrow$ surjective 3.72 $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$
- 3.68 V, W f.d. $S \in \mathcal{L}(V, W)$ $T \in \mathcal{L}(W, V)$ $ST = I$ iff $TS = I$ 3.73 $M(v) = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ s.t. $v = v_1 v_1 + \dots + v_n v_n$
- 3.69 Isomorphism = invertible LM 3.78 V, W f.d. $T \in \mathcal{L}(V, W)$ $\dim \text{range } T = \text{col rank } M(T)$

